Candidate genes for pearl millet flour rancidity

Whole-genome resequencing of a low- and a high-rancidity variety, compared allele by allele



GHB-732 low rancidity — **585** rancidity-related genes

GHB-538 high rancidity — **1,108** rancidity-related genes

▼ both lists merged ▼

1,370

combined genes, compared allele by allele with read-level checks

418 | **257**

validated variety-specific coding alleles

SDP1 LR

TAG lipase · divergent haplotype, 16 missense alleles

Class III POX LR

peroxidase · frameshift p.Ser180fs

Storage oil (TAG) › **DAG / MAG** › **Free fatty acids** › **Peroxides** (H_2O_2, hydroperoxides) › **Off-odour** (hexanal · short shelf life)

HYDROLYSIS

OXIDATION

MAGL HR

MAG lipase · missense p.Glu258Asp

→ **Three candidate genes for RT-qPCR validation**

⚠ **Set aside:** 283 complex loci, including *PgTAGLip3* and LOX3, where apparent loss-of-function calls most likely reflect paralogous or duplicated sequence